Solvency Field Theory

V10.7 Companion Note

Theorem 6b: Branch Selection, Slaved Single-Field Reduction, and Count-Constrained Normalization

Status: Lead 6b branch identified. Minisuperspace branch selection and asymptotic normalization sharpened. Full covariant closure remains open.

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1. Purpose of this note

Theorem 6 asks for the full extension of SFT beyond FLRW minisuperspace: one local covariant action whose reductions reproduce the FLRW count law, the strong-field exact closure, and the perturbative source sector.

The theorem naturally split into three operational fronts:

- **6a**: observer-side bridge narrowing,
- **6b**: full covariant action,
- **6c**: source-sector closure.

V10.5 and V10.6 substantially narrowed 6a and 6c. This note records the first strong triangulation result for **6b**: the best live branch is no longer the original two-field freely kinetic branch. The lead minisuperspace branch is now a **single-field, slaved/constrained branch** in which the source term drives the lag field and the count sector supplies the final normalization.

This note does not claim full covariant closure of 6b. It identifies the correct branch and preserves the strongest current evidence supporting it.

2. What 6a and 6c now force on 6b

Theorems 6a and 6c sharply reduce the freedom inside 6b.

2.1. 6a constraint

The observer-side bridge lives on the screening/completion variable F, not directly on the burden $U = 1 - F$. The best current local completion law is

$$F_{\text{local}}(x) = 1 - e^{-x}, \qquad x = \frac{g}{a_{\text{SFT}}},$$

with first-order bridge structure

$$\delta F \approx +\delta\chi, \qquad \delta\chi = \frac{1}{2\pi}\,\delta\Phi.$$

So the action must admit an exponential completion sector, a screening-side bridge coefficient $\alpha = +1$, and a single-Jacobian bridge coefficient $\lambda = 1/(2\pi)$.

2.2. 6c constraint

The source content is no longer guessed. It is the covariant operation-density / active-mass content

$$\rho_{\text{ops}} = \frac{T_{\mu\nu}u^\mu u^\nu + h^{\mu\nu}T_{\mu\nu}}{h} = \frac{\rho + 3p}{h}$$

in the local clock-field frame, with operation current

$$J^\mu_{\text{ops}} = \rho_{\text{ops}} u^\mu.$$

So the action must couple its lag/bookkeeping scalar to this source content without reintroducing a preferred frame.

These two results already imply that 6b is not the problem of inventing a scalar-tensor theory from nothing. It is the problem of writing the one constrained action whose variations reproduce an exponential screening sector and a u^μ-projected operation-density source sector simultaneously.

3. The original two-field action ansatz

The first serious 6b object was the two-field action ansatz

$$S_{6b} = \int d^4x\,\sqrt{-g}\,\Big[\frac{R}{16\pi G} + \mathcal{L}_\Phi + \mathcal{L}_\chi + \Lambda_C\,\mathcal{C}[g,\Phi,\chi,\psi] + \mathcal{L}_m[g,\psi] - \eta\,\nabla_\mu\chi\,J^\mu_{\text{ops}}\Big],$$

with clock congruence

$$u^\mu = \frac{\nabla^\mu\Phi}{\sqrt{-\nabla_\alpha\Phi\,\nabla^\alpha\Phi}}, \qquad h^{\mu\nu} = g^{\mu\nu} + \nu^\mu\nu^\nu,$$

and scalar sectors of the generic form

$$\mathcal{L}_\Phi = -\frac{K_\Phi}{2}\nabla_\mu\Phi\nabla^\mu\Phi, \qquad \mathcal{L}_\chi = -\frac{K_\chi}{2}\nabla_\mu\chi\nabla^\mu\chi - V(\chi,\Phi).$$

The first bridge-attractor test used the mildest coupled potential consistent with the 6a screening story,

$$V(\chi, \Phi) = V_0 e^{-\chi} + \frac{\mu^2}{2}\left(\chi - \frac{\Phi}{2\pi}\right)^2.$$

The purpose of this test was simple: determine whether the bridge pair $\dot{\chi}/\dot{\Phi} = 1/(2\pi)$ could emerge dynamically while the scalar sector remained background-safe.

4. Why the two-field freely kinetic branch lost

The two-field minisuperspace test did something important: it showed that the bridge ratio can indeed be driven into the right basin. For a broad range of parameters, the late-time ratio

$$\frac{\dot{\chi}}{\dot{\Phi}}$$

settled near

$$\frac{1}{2\pi}.$$

But the same branch generically drove the scalar sector to dominate the Hamiltonian. In reduced units ($8\pi G/3 = 1$), the minisuperspace Hamiltonian was schematically

$$H^2 = \rho + \frac{K_\Phi}{2}\dot{\Phi}^2 + \frac{K_\chi}{2}\dot{\chi}^2 + V(\chi, \Phi),$$

and the numerical branch test showed that once both fields were allowed to propagate freely with ordinary positive kinetic terms, the scalar contribution tended to overwhelm the matter/radiation background.

So the two-field freely kinetic branch failed for a very specific reason:

> It reproduced the kinematic bridge relation only by turning the bookkeeping fields into a runaway energy reservoir.

That is not the right background realization of 6b.

5. The useful simplification: single-field reduction

The bridge-attractor analysis suggested a sharper simplification. Instead of treating Φ and χ as two independent background fields, take the bridge pair as a *definitional relationship*:

$$\Phi = 2\pi\chi.$$

Then automatically

$$\frac{\dot{\chi}}{\dot{\Phi}} = \frac{1}{2\pi},$$

not as a dynamical attractor but as a unit-conversion identity.

This is a real simplification, not a notational trick. If the clock congruence is built from the gradient of the bookkeeping scalar, then

$$\nu^\mu = \frac{\nabla^\mu \Phi}{\sqrt{-\nabla_\alpha \Phi \nabla^\alpha \Phi}} = \frac{2\pi \nabla^\mu \chi}{2\pi \sqrt{-\nabla_\alpha \chi \nabla^\alpha \chi}} = \frac{\nabla^\mu \chi}{\sqrt{-\nabla_\alpha \chi \nabla^\alpha \chi}}.$$

The constant 2π cancels. So the source/current geometry survives the reduction unchanged.

The single-field branch therefore becomes

$$S_{1\mathrm{f}} = \int d^4x \sqrt{-g} \left[\frac{R}{16\pi G} - \frac{K}{2} \nabla_\mu \chi \nabla^\mu \chi - V_0 e^{-\chi} - \eta \, \nabla_\mu \chi \, J^\mu_{\mathrm{ops}} + \mathcal{L}_m + \Lambda_C \, \mathcal{C}[g, \chi, \psi] \right],$$

with

$$\nu^\mu = \frac{\nabla^\mu \chi}{\sqrt{-\nabla_\alpha \chi \nabla^\alpha \chi}}.$$

This branch became worth stress-testing because it removes one unnecessary propagating degree of freedom while preserving the source geometry and the bridge ratio.

6. The source term is a driver, not a tank

In FLRW with lapse $N(t)$, scale factor $a(t)$, homogeneous $\chi(t)$, and perfect-fluid matter with equation of state $p = w\rho$, the operation-density current reduces to

$$J^\mu_{\mathrm{ops}} = \left(\frac{\rho + 3p}{hN}, 0, 0, 0 \right),$$

so the source term becomes

$$L^{\mathrm{FLRW}}_{\mathrm{src}} = -\eta a^3 \dot{\chi} \frac{\rho + 3p}{h}.$$

Two crucial facts follow.

6.1. Lapse independence

The lapse cancels, so

$$\frac{\partial L^{\mathrm{FLRW}}_{\mathrm{src}}}{\partial N} = 0.$$

Therefore the source term is not a direct extra fluid in the Hamiltonian constraint.

6.2. Momentum shift

The conjugate momentum of χ becomes

$$\pi_\chi = a^3 \left(\frac{K}{N} \dot{\chi} - \eta \frac{\rho + 3p}{h} \right).$$

So the source coupling behaves like a momentum shift or driving term, not like a stored background energy density.

This is the cleanest interpretation of the 6c mechanism inside 6b:

> The source term is the fuel line, not the fuel tank.

7. The free single-field stress test

The first single-field minisuperspace stress test kept the same source coupling but varied the kinetic coefficient K.

7.1. Dust

As K was lowered, the background rapidly became well-behaved:

$$\frac{H^2}{\rho} : 4.06 \times 10^3 \longrightarrow 1.000098,$$

and the scalar fraction fell from essentially unity to 10^{-4}-level.

7.2. Radiation

The same behavior occurred more slowly but still decisively:

$$\frac{H^2}{\rho} : 8.66 \times 10^5 \longrightarrow 1.407,$$

with the scalar fraction falling from nearly unity to a modest correction.

7.3. Interpretation

This isolated the real culprit:

> The runaway problem was mostly free kinetic scalar energy, not the source structure itself.

That is a major narrowing result. It means the bookkeeping scalar should probably not propagate like an ordinary extra matter field.

8. The slaved/quasi-static branch

This suggests taking the slaved limit of the single-field model. The sourced scalar equation is

$$K(\ddot{\chi} + 3H\dot{\chi}) - V_0 e^{-\chi} = \eta \nabla_\mu J^\mu_{\text{ops}}.$$

For a perfect fluid with constant w,

$$\nabla_\mu J^\mu_{\text{ops}} = -\frac{3Hw(1+3w)\rho}{h}.$$

Defining

$$A \equiv \frac{3\eta w(1+3w)}{h},$$

the slaved limit $K \to 0$ gives

$$V_0 e^{-\chi} = AH\rho.$$

This defines an effective potential load

$$V_{\text{eff}} = AH\rho.$$

In the unconstrained reduced branch the background equation is

$$H^2 = \rho + V_{\text{eff}} = \rho + AH\rho.$$

8.1. Dust

For dust, $w = 0$, so

$$A = 0.$$

The source driving vanishes and the slaved branch collapses to

$$H^2 = \rho.$$

This is exactly the desired dust behavior.

8.2. Radiation

For radiation, $w = 1/3$, and with $\eta = 1$, $h = 1$,

$$A = 2.$$

So the slaved branch gives a finite correction but no runaway behavior.

This was the first branch that was simultaneously:

- structurally compatible with 6a and 6c,
- background-stable,
- and simple enough to analyze exactly.

9. Count-constrained normalization

The final minisuperspace step was to add a count multiplier β:

$$H^2 = \beta\,(\rho + V_{\text{eff}})\,.$$

The SFT target in reduced units is

$$\beta_{\text{SFT}} = \frac{1}{4\pi^2}.$$

9.1. Dust branch

Since $A = 0$ for dust,

$$H^2 = \beta\rho.$$

So setting

$$\beta = \frac{1}{4\pi^2}$$

reproduces the target dust normalization *exactly*.

9.2. Radiation branch

For radiation, $A = 2$, and the exact reduced equation is

$$H^2 - \beta A\rho H - \beta\rho = 0.$$

Let

$$x(\rho) \equiv \frac{H^2}{\rho}.$$

Then the positive-root solution is

$$\boxed{x(\rho) = \frac{1}{4}\left(\beta A\sqrt{\rho} + \sqrt{\beta^2 A^2 \rho + 4\beta}\right)^2.}$$

In particular,

$$\boxed{\lim_{\rho\to 0}\frac{H^2}{\rho} = \beta.}$$

So if

$$\beta = \frac{1}{4\pi^2},$$

then the count-constrained slaved radiation branch is *asymptotically exact* at the SFT target coefficient.

9.3. Leading correction

Expanding for small ρ gives

$$\frac{H^2}{\rho} = \beta + A\beta^{3/2}\sqrt{\rho} + O(\rho).$$

So the finite-ρ correction dies as $\sqrt{\rho}$.

This matters because the first finite-range numerical average over $a \in [1, 100]$ sat a few percent above the target. The exact asymptotic formula shows that this is just a finite-range averaging effect, not a structural miss.

10. What the asymptotic check proved

The asymptotic radiation check turned the earlier "few percent mismatch" into a solved bookkeeping issue.

With $\beta = 1/(4\pi^2)$,

- pointwise values satisfy $H^2/\rho \to \beta$ exactly as $\rho \to 0$,
- late-time averages converge rapidly toward the same target,
- and the earlier offset is explained by averaging over too-early portions of the radiation branch.

So the count-constrained slaved single-field branch does not merely come close to the SFT coefficient. It approaches it in the mathematically correct way.

11. Updated 6b hierarchy

The current hierarchy is now much sharper.

11.1. Rejected as lead background branch

Two-field freely kinetic branch.

Reason: scalar-energy domination is too severe, even when the bridge ratio is reproduced kinematically.

11.2. Useful but not final

Free single-field branch.

Reason: removes one unnecessary propagating field and dramatically improves the background, but still leaves residual scalar-energy loading when the field is not slaved.

11.3. Promoted to lead 6b subcase

Single-field slaved/constrained branch with count multiplier.

Reason:

- dust branch is exact,
- radiation branch is asymptotically exact,
- the source term remains a driver rather than a stored fluid,
- and the count multiplier supplies the full $4\pi^2$ normalization in the asymptotic background solution.

This is now the best live branch of 6b.

12. Conceptual meaning

The 6b problem is no longer best thought of as "find the right two-scalar field theory." It is better understood as:

- one bookkeeping / lag scalar,
- slaved or constrained rather than freely propagating,
- sourced by the operation-density current,
- with the count sector fixing the final normalization.

Equivalently:

- 6a fixes the bridge structure,
- 6c fixes the source content,
- the count sector fixes the coefficient.

That is the sharpest triangulation of 6b obtained so far.

13. What remains open

This note does not close 6b. It clarifies where the remaining work lives.

Three major items remain:

1. **Full local covariant count constraint.** The minisuperspace count multiplier must be lifted to the true local covariant count relation.

2. **Fully covariant single-field action.** The slaved branch must be written as a local covariant action or constrained action, not only as a minisuperspace reduction.

3. **Strong-field and perturbative reductions.** The single-field constrained branch must still reproduce the strong-field exponential metric sector and the perturbative/shatter sector in the correct limits.

But the branch structure itself is now much clearer than before.

14. Bottom line

The strongest current 6b statement is:

> **The best live 6b branch is a single-field, slaved/constrained branch in which the source term drives the lag field and the count multiplier supplies the $4\pi^2$ normalization. In this branch dust is exact and radiation is asymptotically exact at the SFT target coefficient.**

This does not finish 6b. But it selects the correct branch and sharply identifies where the remaining work now lives. That is enough to promote the result to a V10.7-level companion note.